Lab08

Mateus Bortolucci

```
library(reticulate)
py_require(c("polars", "gdown"))
repl_python()
```

```
Python 3.13.16 (/cloud/home/r3941873/.cache/R/reticulate/uv/cache/archive-v0/
ffYMB6WOrE3CDBpX/bin/python)
Reticulate 1.47.0 REPL -- A Python interpreter in R.
Enter 'exit' or 'quit' to exit the REPL and return to R.
```

```
exit
```

```
import gzip, shutil

for nome in ["title.basics0", "title.ratings", "title.principals0"]:
    with gzip.open(f"{nome}.tsv.gz", "rb") as f_in, open(f"{nome}.tsv", "wb")
as f_out:
        shutil.copyfileobj(f_in, f_out)

import polars as pl

OPC = dict(separator="\t", null_values=["\\N"], quote_char=None,
infer_schema=False)

basics = (
    pl.scan_csv("title.basics0.tsv", **OPC)
    .filter(pl.col("titleType") == "movie")
    .select("tconst", "primaryTitle", "genres")
)

ratings = (
    pl.scan_csv("title.ratings.tsv", **OPC)
    .select(
        "tconst",
        pl.col("averageRating").cast(pl.Float64),
        pl.col("numVotes").cast(pl.Int64),
    )
```

```
)

principals = (
    pl.scan_csv("title.principals0.tsv", **OPC)
    .select("tconst", "nconst", "category")
)

# Resultado pequeno: pode ficar em memória
filmes = basics.join(ratings, on="tconst",
how="inner").collect(engine="streaming")

# Q1
q1 = (
    filmes.sort(["averageRating", "numVotes", "primaryTitle"],
                descending=[True, True, False])
    .head(5)
)

# Q2
q2 = (
    filmes.filter(pl.col("averageRating") > 8)
    .drop_nulls("genres")
    .with_columns(pl.col("genres").str.split(","))
    .explode("genres")
    .group_by("genres")
    .agg(pl.len().alias("n_filmes"))
    .sort(["n_filmes", "genres"], descending=[True, False])
)


# Q3
bons_filmes = filmes.filter(pl.col("averageRating") >
7.5).select("tconst").lazy()

q3 = (
    principals
    .filter(pl.col("category").is_in(["actor", "actress"]))
    .join(bons_filmes, on="tconst", how="inner")
    .select("nconst", "tconst").unique()
    .group_by("nconst")
    .agg(pl.len().alias("n_filmes"))
    .sort(["n_filmes", "nconst"], descending=[True, False])
    .head(3)
    .collect(engine="streaming")
)
```